

Cross-Lingual Cognitive Synergy for Constrained Humor Generation in Large Language Models

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Problem Statement & Motivation

- Humor is a very important element in human social interaction
- AI advances has increased its deployment in socially interactive environments
- LLM's ability to generate humor remain limited
 - **Reason:** LLM training object conflicts with the requirements for humor generation
- Existing research are less focused on humor generation, with emphasis on detection



CLEF 2026 JOKER Task 4: Humor Generation

Description

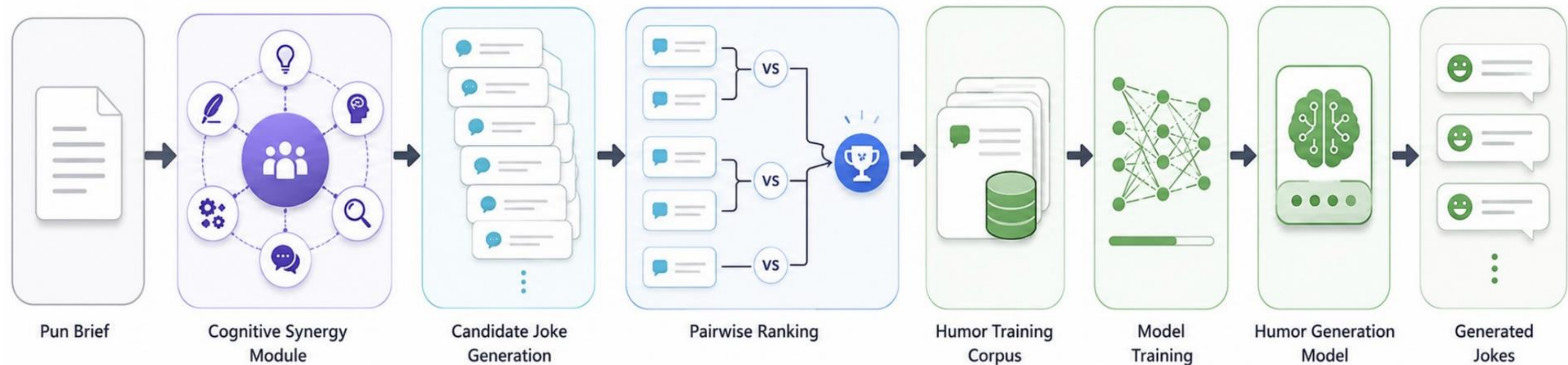
Given a **structured brief about a pun** (the pun word and two distinct senses that the wordplay should activate, sometimes with supporting keywords), the system must generate a short, funny, natural-sounding humorous text in the target language.

A good generation:

1. Uses the given pun word (or a clear morphological variant).
2. Activates both senses – the joke should not work if either sense is removed.
3. Reads naturally – judges down-weight forced or template-y phrasing.
4. Is in the target language.



Methodology - Architecture Diagram



Architecture Diagram - Cognitive Synergy Framework for Crosslingual Humor Generation

Cognitive Synergy Framework

In Cognitive Synergy Framework, we define six Cognitive Personas, each grounded in a specific psychological theory of humor

Persona	Humor Theory	Mechanism	Cognitive Focus
Neurotic Cynic Observer	Relief Theory	Tension Release	Internal anxiety, overthinking, and social insecurity.
	Superiority Theory	Social Critique	Hypocrisy, biting sarcasm, and moral contradictions.
	Incongruity	Social Mapping	Mundane minutiae and unwritten awkward social norms.
Wordsmith Optimist	Linguistic Benign Violation	Ambiguity Recontextualization	Puns, double entendres, and phonological play. Wholesome misinterpretations of potentially negative traits.
Absurdist	Incongruity	Surrealism	Non-sequiturs, dream logic, and fractured causality.

Cognitive Synergy Framework Personas (Source: HumorGen Paper - EMNLP 2026)

Experimental Setup - Overview

Question:

How much impact does CSF Guided data generating impact humor generation in Large language Models

	Kimi-CSF	Kimi-Plain	JOKER-32B	JOKER-14B
<i>Training stage</i>				
CSF on training briefs	–	–	✓	✓
Pairwise ranking on train	–	–	✓	✓
SFT on top-ranked candidates	–	–	✓	✓
<i>Test stage</i>				
CSF at test time	✓	–	–	–
Pairwise ranking at test time	✓	–	–	–
Generation backbone	Kimi K2.6	Kimi K2.6	Qwen3-32B	Qwen3-14B
Constraint verification	✓	✓	✓	✓
Trained on organizer reference jokes	–	–	–	–

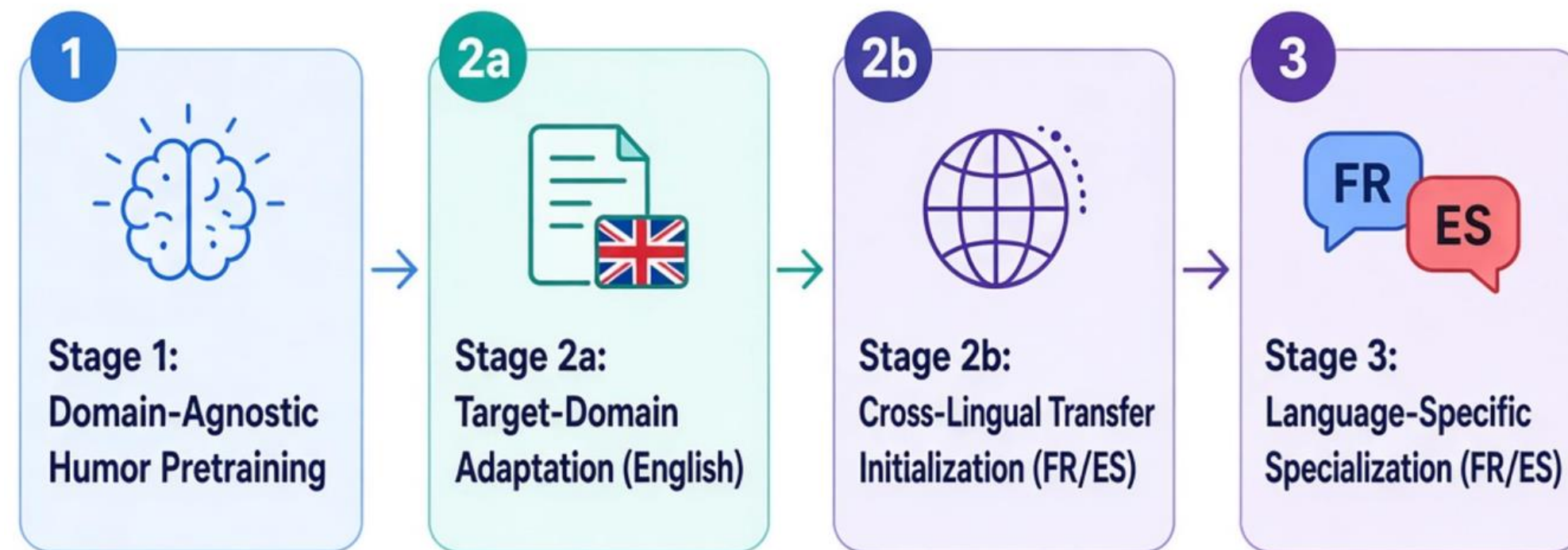
Experimental Setup - Dataset & Model Training

Datasets:

- CSF Generated Data on SEMEVAL Humor generation benchmark task news-headline based train set(1200 * 10 Synthetic Humor generation data)
- CSF generated data on CLEF 2026 Task 4 Training Dataset including English monolingual (3,985), French monolingual (3,952), Spanish monolingual (3,101), Multilingual (EN+FR+ES) (11,038)

Model Training steps

We demonstrated the different experiment performed on our model training for the humor generation models in the three languages



Model Training Set-up

Experimental Setup - Generation Prompt

Write a short, natural, very funny joke.

The humor should arise from the interplay between the two meanings of “{pun_word}” below.

Pun word : {pun_word}

Meaning 1 : {sense_a}

Meaning 2 : {sense_b}

Keywords : {keywords}

The joke should feel natural rather than explanatory.

Avoid simply listing or defining the meanings.

Use surprise, misdirection, playful reinterpretation, or an unexpected connection between the two meanings.

Do NOT explain the pun or its double meaning — let the wordplay speak for itself.

Requirements:

- Use “{pun_word}” (or a clear inflection) in the joke
- Rely on both meanings of “{pun_word}”
- Write entirely in {Language}
- Keep it short and punchy
- Output only the joke

Joke Generation Prompt

Results - Sample Joke Generation

We show below sample jokes from our model from the task accross the three languages (English(EN), French (FR) and Spanish (ES)).

English het_1261

INPUT

Pun word: *vein*

Sense A: blood vessel toward heart

Sense B: to no avail (*in vain*)

Keywords: I.V.; in vein; all in vein; save; tried to save him; vein

OUTPUT (JOKE)

*"He paid the hospital twelve grand for an I.V., watched the bag drip empty into his arm, then flatlined anyway. Turns out the whole transfusion was in **vein**."*

Français fr_1009

INPUT

Pun word: *sortir de ses gonds*

Sense A: sortir de ses gonds (porte, sens propre)

Sense B: sortir de ses gonds = s'énerver (sens figuré)

OUTPUT (JOKE)

*"Deux menuisiers installaient des portes, rien ne s'alignait. Plus ça forçait, plus ça coïncait : ils finissent juste par **sortir de leurs gonds**."*

Español es_0780

INPUT

Pun word: *facultades*

Sense A: facultades (universidad)

Sense B: facultades (capacidades mentales/físicas)

OUTPUT (JOKE)

*"Estudié diez años en la facultad de Medicina. Salí con título, con deuda... y sin **facultades**."*

Generated Joke Samples

Results - JokerLab Arena

We report our models performance accross the three languages (English(EN), French (FR) and Spanish (ES) on the JOKER Task 4 Leaderboard

System	EN		FR		ES	
	Rank	BT	Rank	BT	Rank	BT
Kimi-CSF	1	1257	1	1395	1	1515
Kimi-Plain	2	1225	11	1272	2	1437
HumorGen-JOKER-32B	4	1135	18	1246	3	1369
HumorGen-JOKER-14B	3	1168	16	1256	5	1329

Model Rankings on the Joker Arena Leaderboard

Observation:

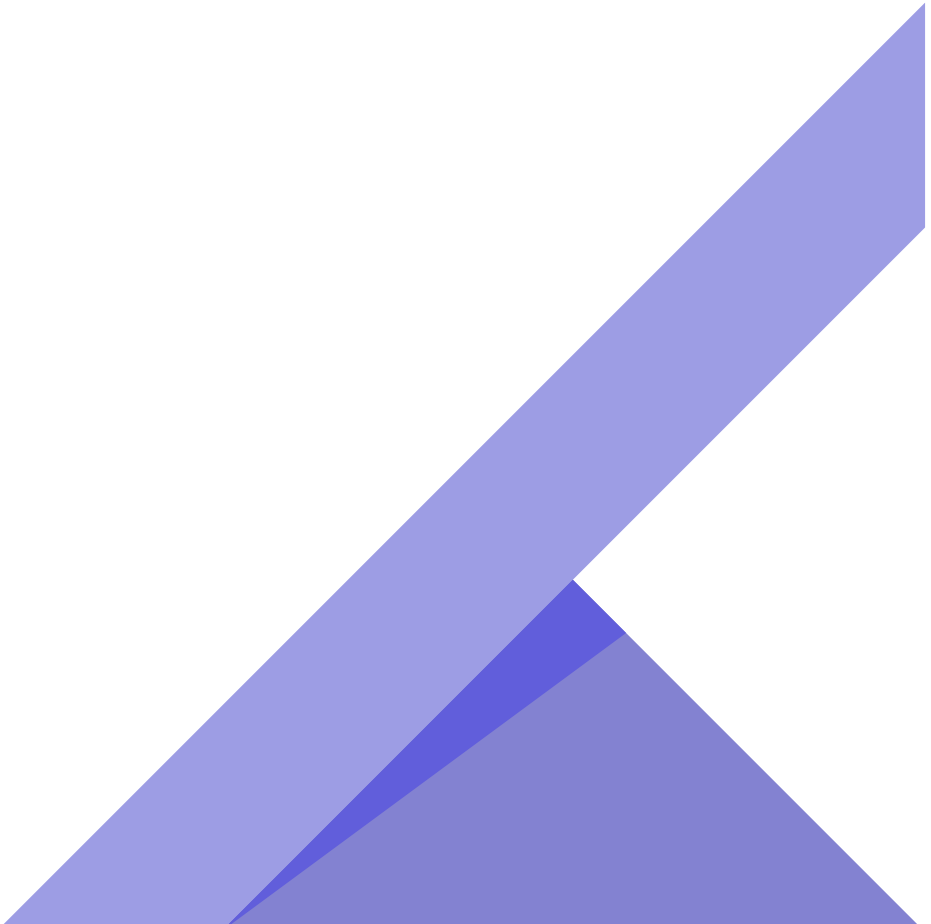
- KIMI-CSF won all 3 languages
- HumorGen-JOKER models achieved top positions

Conclusion

Findings

- Guided creativity is the dominant factor in humor generation beyond prompting techniques and other approaches
- Student models trained using the CSF generated synthetic data almost matches teacher model ability in humor generation

Future Directions

- Generalizable approaches to multimodal humor generation, including audio and images
 - Expanding language coverage for textual humor generation
 - Developing culturally grounded approaches to humor generation
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Thank You

Read More here



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